RELATIVE MOTION

LEARNING GOALS

Students will:

- Determine the velocity of objects in different frames of reference

PREPARATION AT HOME

Videos

- Train Station – What is moving?

Activities

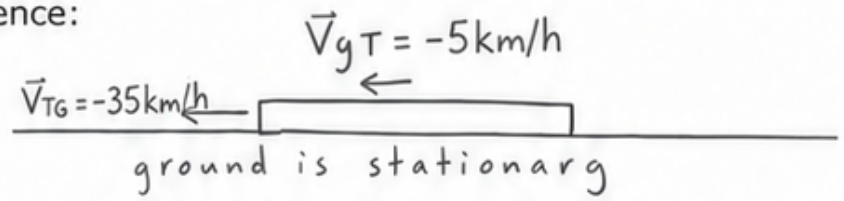
- Physlets – Try Relative Velocity in 1D and 2D

Homework Quiz

MOTION IN 1-D

Imagine you are sitting on a train. The train is traveling at 35 km/h [W]. You stand up and walk forward along the aisle of the train at 5 km/h. In this problem there are essentially three points of view or frames of reference:

1. the train
2. the ground
3. you



W(−) ⟷ E(+)

When we describe motion, we say WHAT is moving and WITH RESPECT TO some point of view or frame of reference.

1. Consider the following statements:
   - The velocity of the train with respect to the ground is 35 km/h [W]
   - Your velocity with respect to the train is 5 km/h [W]
   - Your velocity with respect to the ground is 40 km/h [W]
   - The velocity of the ground with respect to you is 35 km/h [E]
   - The velocity of the ground with respect to the train is 35 km/h [E]

   One of the above statements is NOT correct. Which one is wrong? Explain why.

   Statement 4 is incorrect because your velocity adds with the velocity of the train, so it should be 40kmh (E), The magnitude is wrong. $\vec{V}_{yG} = 40km/h(W)$ $\vec{V}_{Gy} = 40kmh(E]$

2. You are on the same train as above. This time you walk towards the back of the train at 5 km/h along the aisle. Answer the following:

   a) The velocity of the train with respect to the ground is $-35kmh$

   b) Your velocity with respect to the train is $+5kmh$

   c) Your velocity with respect to the ground is $-30kmh$

   d) The velocity of the ground with respect to you is $+30kmh$

   e) The velocity of the ground with respect to the train is $+35kmh$

NOTATION

We need a short, mathematical way of making the same statements as above.

If some object A has a velocity with respect to some other object or frame of reference B, we use the notation

$$\vec{v}_{AB}$$

This is read as the "velocity of A with respect to B". The first subscript is the object, the second subscript is the frame of reference.

The order of the subscripts is IMPORTANT. In the example we started with, the train has a velocity with respect to the ground of 35 km/h [W]. If you are standing on the ground watching the train go by, this is what you see. **V_{TG} = 35 km/h [W]**

BUT if you are standing on the train and you look out the window, the ground appears to be going by you in the opposite direction. **V_{GT} = 35 km/h [E]**

4. Take each of the statements in question 2 above and rewrite them using the notation given above.

a) $\vec{V}_{Tg} = 35\ km/h\ [W]$

b) $\vec{V}_{y+} = 5\ km/h\ [E]$

c) $\vec{V}_{yg} = 30\ km/h\ [W]$

d) $\vec{V}_{gy} = 30\ km/h\ [E]$

e) $\vec{V}_{gT} = 35\ km/h\ [E]$

CROSSING THE RIVER

Say you are in a boat and you need to cross a river. You know that the boat's motor can put out a steady speed of 10. m/s but there is a current of 3.0 m/s. You have two options – aim a bit **upstream** to compensate for the current (Fig. 1) or point the boat straight across and end up a little **downstream** of your start (Fig. 2).

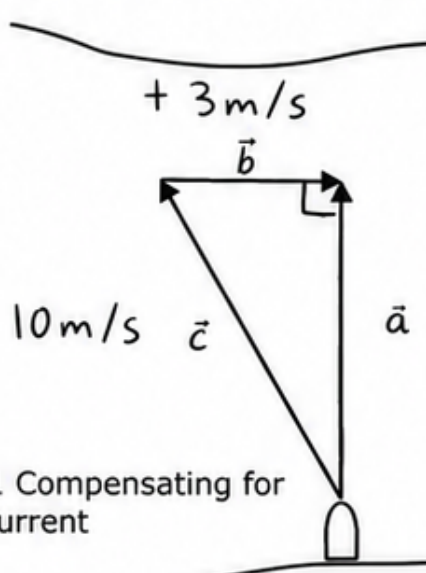


Fig. 1 Compensating for the current

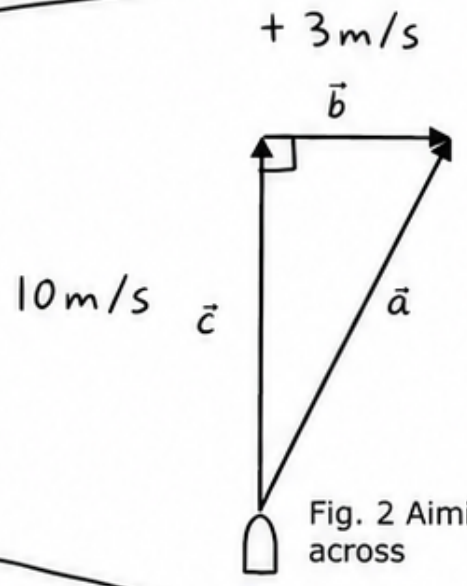
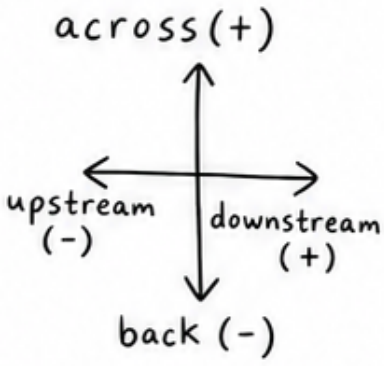


Fig. 2 Aiming straight across



a) Make a quick prediction – which trip will take the shortest time to reach the opposite side of the river? Circle your choice.

Fig. 1 (Fig. 2) Both the Same

b) In the diagrams, which two vectors are added together and which is the resultant?

$\vec{b}$ and $\vec{c}$ are added and $\vec{a}$ is the resaltant

c) Label the speeds of 10. m/s and 3.0 m/s on each diagram.

d) Identify the frame of reference for each velocity vector shown:

Velocity	*Vector Notation*	*Vector $\vec{a}$, $\vec{b}$, or $\vec{c}$?*
Velocity of the boat with respect to the ground	$\vec{v}_{BG}$	$\vec{a}$
Velocity of the boat with respect to the water	$\vec{v}_{BW}$	$\vec{c}$
Velocity of the water with respect to the ground	$\vec{v}_{WG}$	$\vec{b}$

Check your answers on the board or with the teacher! (or on karolsohes kassil.com)

e) Calculate the velocity of the boat with respect to the ground in both situations and determine which will cross the river first.

1

$a^2 + b^2 = c^2$

$a^2 + 3^2 = 10^2$

$a = \sqrt{91}$

$a = 9.5\ m/s$

$\vec{a} = 9.5\ m/s$ [across]

let d equal distance

$\Delta t = d / v$

$\Delta t_1 = d / 9.5$

2

$3^2 + 10^2 = a^2$

$a^2 = 109$

$a = \sqrt{109}$

$a = 10.44$

find θ

$\theta = \sin^{-1}\left(\frac{b}{a}\right)$

$\theta = \sin^{-1}\left(\frac{3}{10.44}\right)$

$\theta = 17°$ downstream of across

$\vec{a} = 10\ m/s$ [across 17° downstream]

let d equal distance

$\Delta t = d / v$

$\Delta t_2 = d / 10$

$\Delta t_1 > \Delta t_2$

∴ boat 2 crosses the river first

A NOTE ON SUBSCRIPTS

Notice that with both of the diagrams in the river situation, you could write the following vector addition statement:

Starts with B, Ends with G

Starts with B
Ends with G → $\vec{v}_{BG} = \vec{v}_{BW} + \vec{v}_{WG}$

Must be the same

If your vector diagram is correct, then you will always see this pattern.

DETERMINING FRAMES OF REFERENCE

You will often analyze the motion of boats or swimmers or planes. In these situations, the velocity of the object with respect to the **water/air** is in the direction that they aim themselves. The terms **airspeed** and **heading** refer to velocity with respect to air.

For swimmers or boats, whether the swimmer is in moving or still water, her velocity with respect to the water is the same. This is her swimming ability. Similarly, the plane's velocity with respect to the air is like the velocity of the swimmer with respect to the water.

The velocity of an object with respect to the **ground** is where they actually go. It is what you would see if you were watching from the ground.

Note that a west wind means wind that is coming **from** the west. A north wind is blowing **from** the north. Also, the term **upstream** means in the direction opposite to the current and **downstream** means with the current as in the river example above.

DIAGRAMS OF RELATIVE VELOCITIES

For each of the examples given below,

- ☐ draw the vector diagram of the velocities
- ☐ **label** each vector using the correct notation and the values given (speeds and directions)
- ☐ write the vector addition statement that describes the diagram (must match the pattern)

1. A new pilot forgets to compensate for the wind. He believes he is heading [N 20° W] at 250 km/h but the he's in a north wind with a velocity of 60.0 km/h. He flies for 30 minutes.

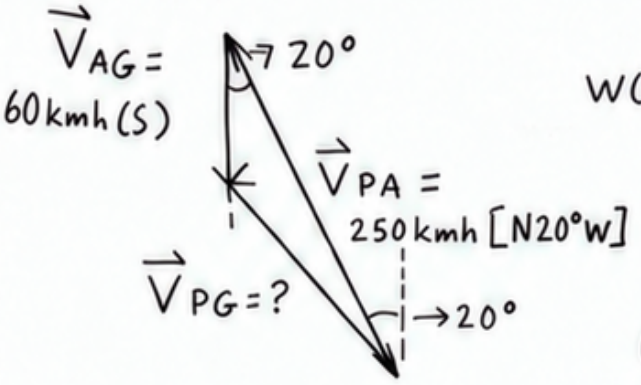
2. An ocean liner is travelling 18 km/h due south. A passenger on the deck walks toward the rear of the ship at 3.0 m/s.

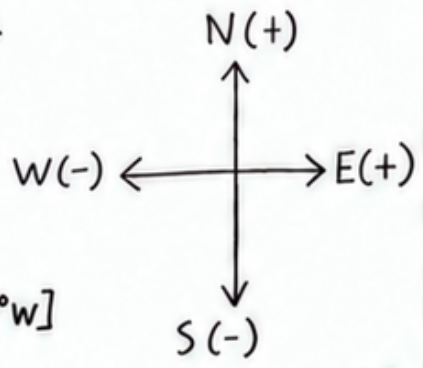
3. The same ship as in #2, but the passenger has turned right and is walking towards the railing.

4. A plane flies in wind of velocity [65 km/h [S30°E]. After 4.0 hours, it arrives at its location which is 1200 km [E15°N] of its starting point.

Homework: solve each of the situations above for the unknown vector.

① $\vec{V}_{PA} + \vec{V}_{AG} = \vec{V}_{PG}$





$\theta = \tan^{-1}\left(\frac{174.9}{85.5}\right)$

$\theta = 64°$ north of west

$\vec{V}_{AGx} = 0$ km/h

$\vec{V}_{AGy} = -60$ kmh

$\vec{V}_{PAx} = -\sin 20° \times 250$

$\vec{V}_{PAx} = -85.5$ kmh

$\vec{V}_{PAy} = \cos 20° \times 250$

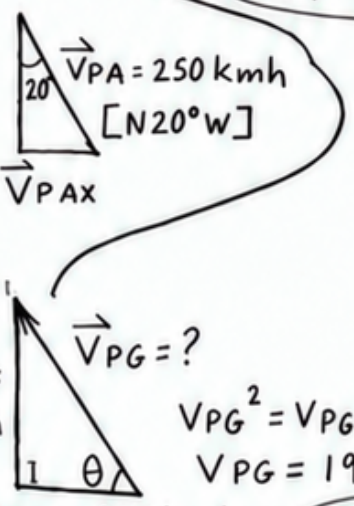
$\vec{V}_{PAy} = +234.9$ kmh

$\vec{V}_{PGx} = \vec{V}_{PAx}$

$\vec{V}_{PGx} = -85.5$ kmh

$\vec{V}_{PGy} = \vec{V}_{PAy} + \vec{V}_{AGy}$

$\vec{V}_{PGy} = +174.9$ kmh



$V_{PG}^2 = V_{PGx}^2 + V_{PGy}^2$

$V_{PG} = 194.7$ kmh

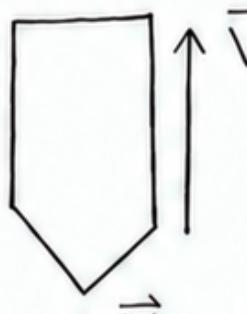
$V_{PG} = 1.9 \times 10^2$ kmh

$d = v \times at$

$d = 97$ km

$\therefore$ 97 km [W64° N]

② N(+) S(-)



$\vec{V}_{SG} = 18$ kmh (S)

$\vec{V}_{SG} = 18/3.6$ m/s [S]

$\vec{V}_{SG} = 5$ m/s [S]

$\vec{V}_{PG} = \vec{V}_{SG} + \vec{V}_{PS}$

$\vec{V}_{PG} = -5 + 3$

$V_{PG} = -2$ m/s

$V_{PG} = 2.0$ m/s [S]

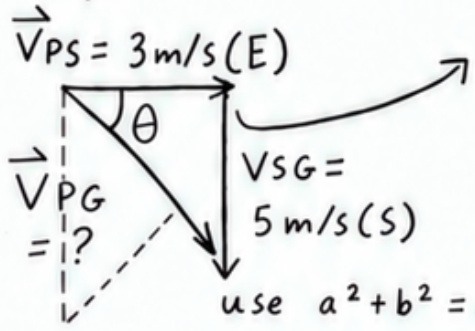
③ $\vec{V}_{PG} = \vec{V}_{PS} + \vec{V}_{SG}$

N(+) S(-) W(-) E(+)



$\vec{V}_{PS} = 3$ m/s (E)

$\vec{V}_{SG} = 5$ m/s (S)



$\theta = \tan^{-1}\left(\frac{5}{3}\right)$

$\theta = 59°$ south of east

use $a^2 + b^2 = c^2$

$V_{PG} = \sqrt{V_{SG}^2 + V_{PS}^2}$

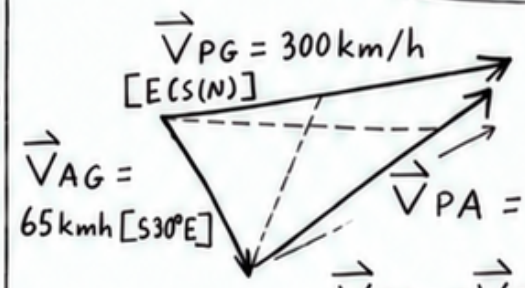
$V_{PG} = 5.8$ m/s

$\vec{V}_{PG} = 5.8$ m/s [E 59° S]

④ $d = 1200$ km $V = \frac{1200}{4}$

$\Delta t = 4$ hrs

$V = 300$ km/h



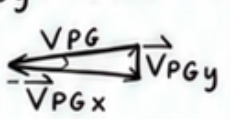


$\vec{V}_{AGx} = \sin 30 \vec{V}_{AG}$

$\vec{V}_{AGx} = +32.5$ km/h

$\vec{V}_{AGy} = \cos 30 \vec{V}_{AG}$

$\vec{V}_{AGy} = -56.3$ km/h



$\vec{V}_{PGx} = \cos 15 \vec{V}_{PG}$

$\vec{V}_{PGx} = +289.8$ km/h

$V_{PGy} = \sin 15' \vec{V}_{PG}$

$V_{PGy} = +77.6$ km/h

$\vec{V}_{PAx} = \vec{V}_{PGx} - \vec{V}_{AGx}$

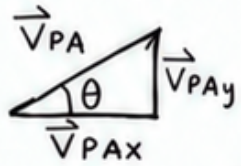
$\vec{V}_{PAx} = 289.8 - 32.5$

$V_{PAx} = +257.3$ km/h

$\vec{V}_{PAy} = \vec{V}_{PGy} - \vec{V}_{AGy}$

$V_{PAy} = 77.6 + 56.3$

$V_{PAy} = +133.9$ km/h



use $a^2 + b^2 = c^2$

$V_{PA} = \sqrt{V_{PAy}^2 + V_{PAx}^2}$

$V_{PA} = 290$ km/h

$\theta = \tan^{-1}\left(\frac{V_{PAy}}{V_{PAx}}\right)$

$\theta = 27°$ north of east

$\therefore \vec{V}_{PA} = 2.9 \times 10^2$ kmh [E 27° N]